

Tuning DALS: An 80-Trial Hyperparameter Search

Can the level-set hyperparameters move the needle, or is the ceiling elsewhere?

CS269 Project — Deformable Models

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1 Question

DALS couples a U-Net trunk to a differentiable Chan–Vese level-set evolution with learnable parameters $(\mu, \lambda_1, \lambda_2, dt)$, unrolled over n_{iter} steps. The structural and loss hyperparameters were never searched — they sit at hand-picked defaults. This mini report asks a simple question: *does any hyperparameter configuration beat the current DALs baselines (FLAME-1 IoU 0.747, FLAME-3 IoU 0.122) beyond training noise?* A second question rides along: does optimizing for *boundary* fidelity select a different configuration than optimizing for *region* overlap?

2 Method

Random search, four arms = $\{\text{FLAME-3, FLAME-1}\} \times \{\text{val-IoU objective, val-BF objective}\}$, 20 trials each (80 trials total). Each trial trains DALs at reduced epochs (FLAME-3: 25; FLAME-1: 12 at 1024px), records *both* val IoU and val boundary F-score@2px, and the arm ranks by its own objective. The winner of each arm is retrained at the full 50 epochs and scored on the held-out test split (94 FLAME-3 / 301 FLAME-1 frames).

Search space. $n_{\text{iter}} \in \{3, 5, 8, 12\}$; $\mu \in [0.1, 1.0]$; $\lambda_1, \lambda_2 \in [0.5, 2.0]$; $dt \in [0.05, 0.3]$; $\text{loss} \in \{\text{bce_dice, focal_tversky}\}$ with Focal–Tversky $\beta \in [0.5, 0.8]$ ($\alpha = 1 - \beta$), $\gamma \in [0.5, 1.0]$. Fixed at defaults: $\text{lr} = 10^{-4}$, $\text{batch} = 4$, light augmentation. Config sampling is seeded for reproducibility.

3 Results

Every one of the four arms ranked the *same* configuration first:

$$n_{\text{iter}}=3, \mu=0.21, \lambda_1=0.93, \lambda_2=1.38, dt=0.19, \text{loss} = \text{bce_dice}.$$

Table 1: Winning config retrained at 50 epochs, test-split evaluation. Baseline is the existing DALs checkpoint.

Dataset	Search objective	Baseline IoU	Tuned test IoU	Tuned test Dice
FLAME-3	val IoU	0.122	0.120	0.184
FLAME-3	val BF	0.122	0.129	0.193
FLAME-1	val IoU	0.747	0.753	0.858
FLAME-1	val BF	0.747	0.745	0.853

4 Findings

No region-vs-boundary tradeoff. On both datasets the val-IoU arm and the val-BF arm selected the *identical* winning hyperparameters. Optimizing for boundary fidelity does not pull the DALs configuration anywhere different than optimizing for region overlap — there is no Pareto frontier to exploit.

FLAME-3 is a wash. The tuned test IoU (0.120 / 0.129) straddles the 0.122 baseline. Tellingly, two retrains of the *same* config differed by 0.009 IoU from training stochasticity alone (no fixed retrain seed), so the run-to-run noise floor is $\approx \pm 0.01$ — the entire tuned-vs-baseline gap sits inside it. This is consistent with the detection-bottleneck story: on FLAME-3 the ground truth is hidden $\geq 150^\circ\text{C}$ thermal support, and the ceiling is set by what the U-Net trunk can *detect*, not by the level-set hyperparameters.

FLAME-1 is a wash too, but confirms a strong baseline. Best tuned test IoU 0.753 vs. 0.747 baseline is a +0.006 nudge, again within noise. Tuning neither broke nor meaningfully improved the already-strong visible-flame regime.

$n_{\text{iter}} = 3$ and bce_dice dominate. Across all arms the top trials clustered on few level-set iterations and the bce_dice loss; high- n_{iter} and Focal-Tversky trials were frequently unstable (several near-zero val IoU). The default DALs recipe is already close to optimal.

5 Bottom line

The 80-trial search found no hyperparameter configuration that beats the existing DALs baselines beyond training noise on either dataset. The defaults are effectively already tuned. If the goal is to actually move the FLAME-3 number, the lever is the detection trunk — recovering more of the hidden thermal support — not the contour hyperparameters.